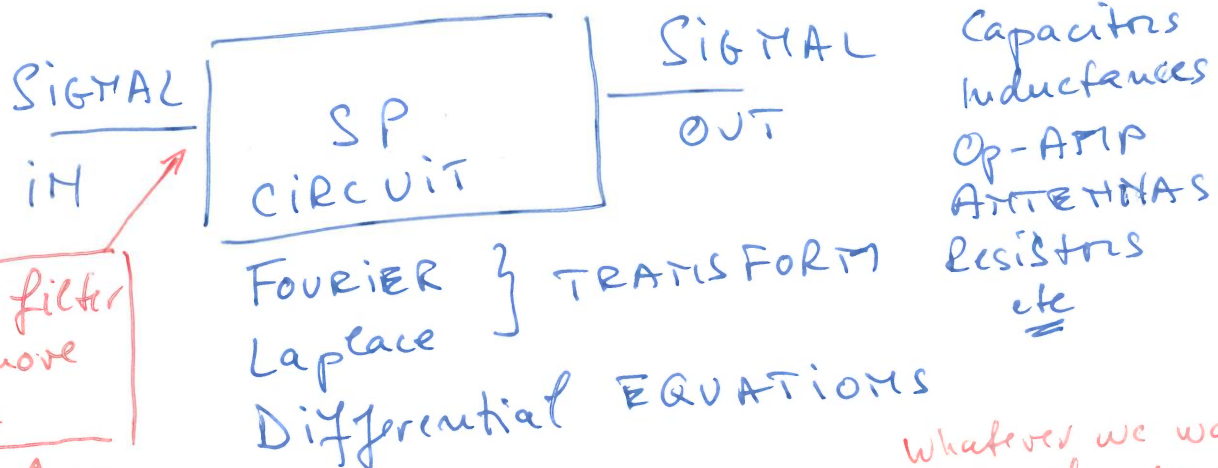


Introduction

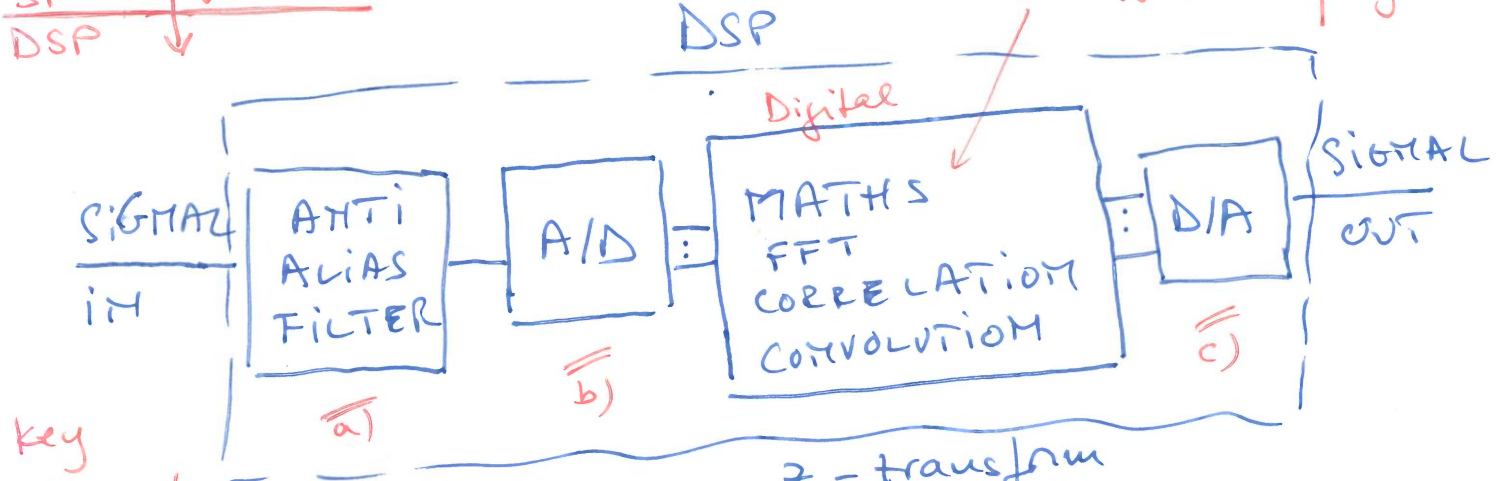
SP vs DSP
 Signal Processing vs Digital Signal Processing



Input filter to remove noise

SP vs DSP

Whatever we want to do (program)



key elements
 a) b) c)
 Analogue

z-transform
 Difference Equations
 MULTIPLY-AND-ADD
 (Recurrence relations)
 MATHS + PROGRAMMING

Example

SP

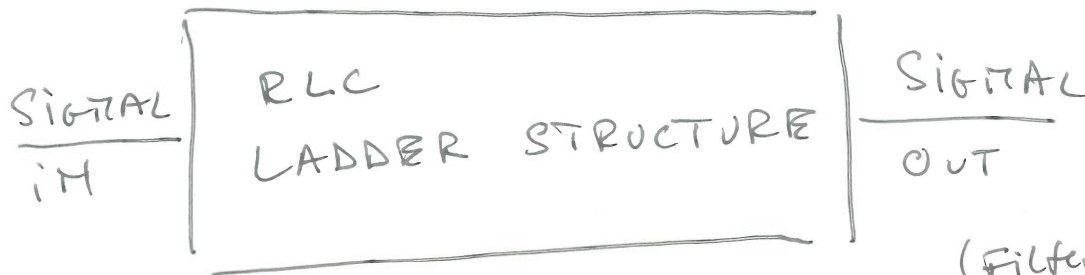
6th order bandpass chebychev filter

(2)

Faster

change filter?

change circuit



Toby will tell you how-to (Filter) (tables)
(Parameters are component values
Some maths needed for
baseband $\rightarrow$ bandpass
scaling)

No explicit maths in the circuit!
Component values critical!

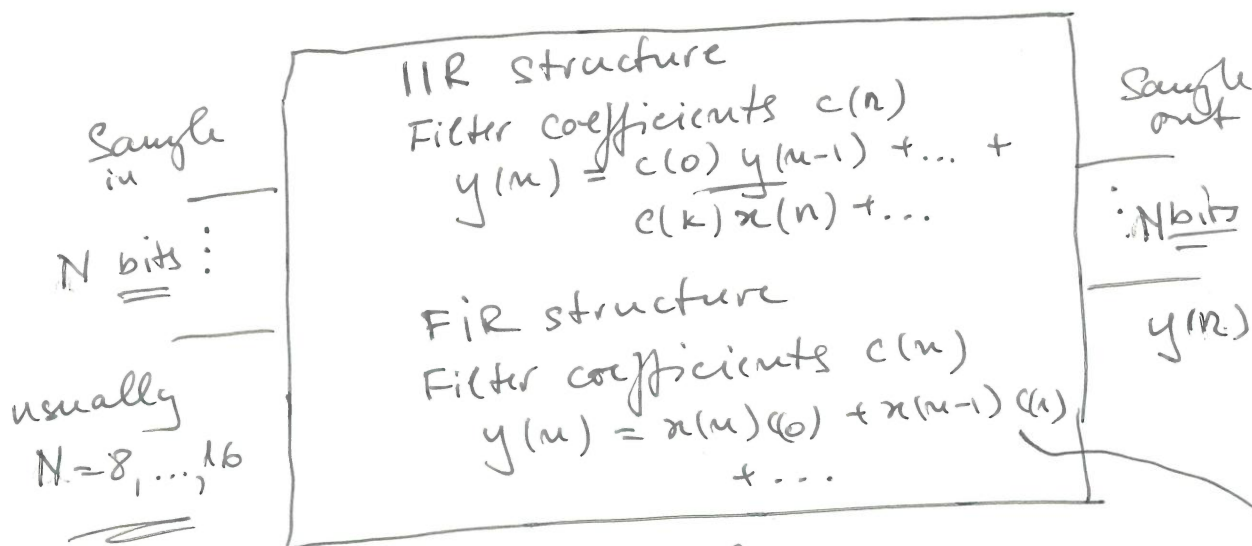
DSP

Slower

change filter?

change coefficient

More filter options



usually $N=8, \dots, 16$
 $x(n)$ $\rightarrow$ N bit $\rightarrow$ $\rightarrow$ Numerical issues
 $\rightarrow$ Speed issues (processing power)
FFT could be used instead
to implement convolution

e.g. Fiiir page, Matlab $\rightarrow$ coefficients

MODULE STRUCTURE

(3)

80% EXAM - Plan 3 out of 4 Qs
ALIASING } Calculate
FIR } Theory

Focus on EXAMPLES
in CLASS

Might be different in
structure / coverage
from past questions

{ 10% VIVA → DEMO in LAB (Group^{1/2})
10% REPORT → Short report (individual)
→ Need to pass each (no no shows please)

PRACTICALS (2W) - 10X + ECTS

Focus on ST board

- Template code provided
- Analyse existing setups
 - Sampling
 - Frequency transfer function
- Modify code to get your own filter
 - determine filter coeff - INT
 - FIR page on other
 - Matlab support provided (EMMAHUEL)
- Anti aliasing
- Investigation / demo / Report

First comment on the sampling theorem:

(5)

The sampling theorem is not obviously absurd.

Easy to use, hard to understand.

Hidden assumptions

→ regular sampling (stay with it)
↳ vs irregular

→ no jitter or other imperfections

→ Finite spectrum

Still, we now can answer the following

Q1

Question: What is the minimum

sampling frequency if $F_{MAX} = 10 \text{ kHz}$

Answer

No overlap good

$$\text{So } F_{MAX} < F_s - F_{MAX}$$

$$\Rightarrow F_s > 2 F_{MAX} = 20 \text{ kHz}$$

(Nyquist)

Look up: Faster-than-Nyquist

Subjects

Spent enough time
here: 1)

(4)

- 1) Sampling, aliasing, quantisation
→ anti-alias filter (analogue)
→ sample & hold, reconstruction of
order 1 (hold)
→ imaging (related to alias)

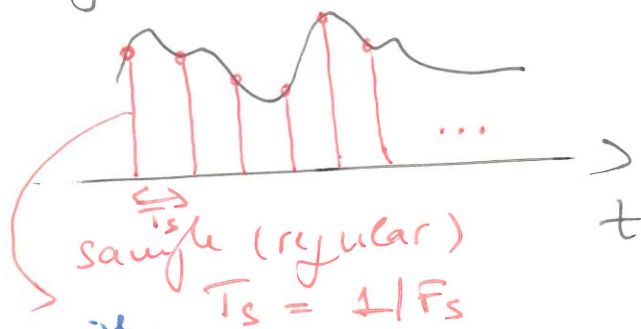
Need Frequency representation

Bandwidth, spectrum

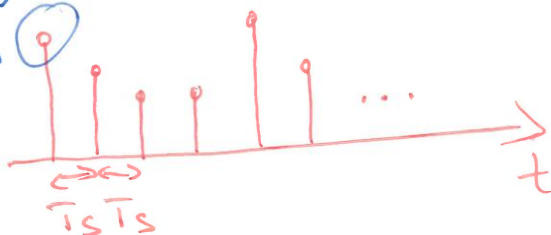
Why?

Simplest way to understand
Sampling

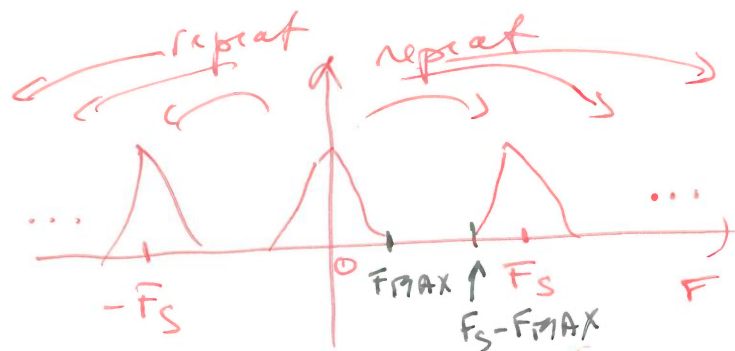
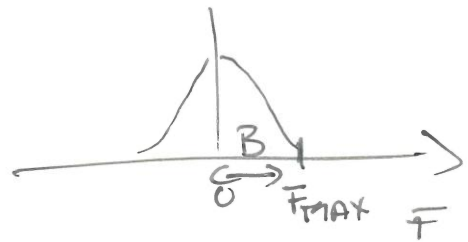
Signal in time



infinite
precision
number



Signal in Frequency



F_s sampling frequency
 T_s sampling period

Simple!

Repeat

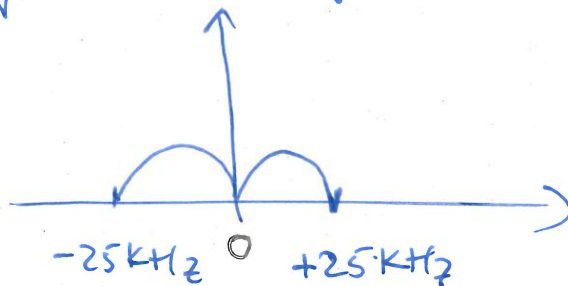
Overlap → Alias

Q2

(6)

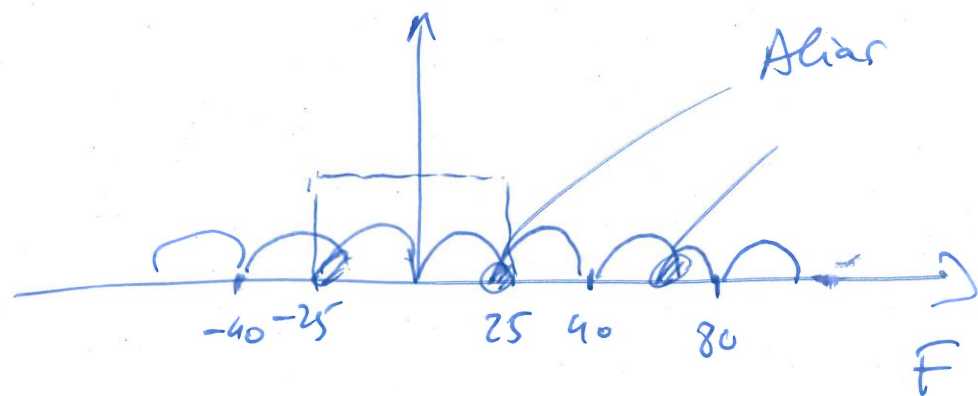
A signal with maximum frequency $F_{MAX} = 25 \text{ kHz}$ is sampled at a rate $F_S = 40 \text{ kHz}$.

- Determine if there will be aliasing
- If the signal looks in frequency as below, sketch the spectrum of the sampled signal



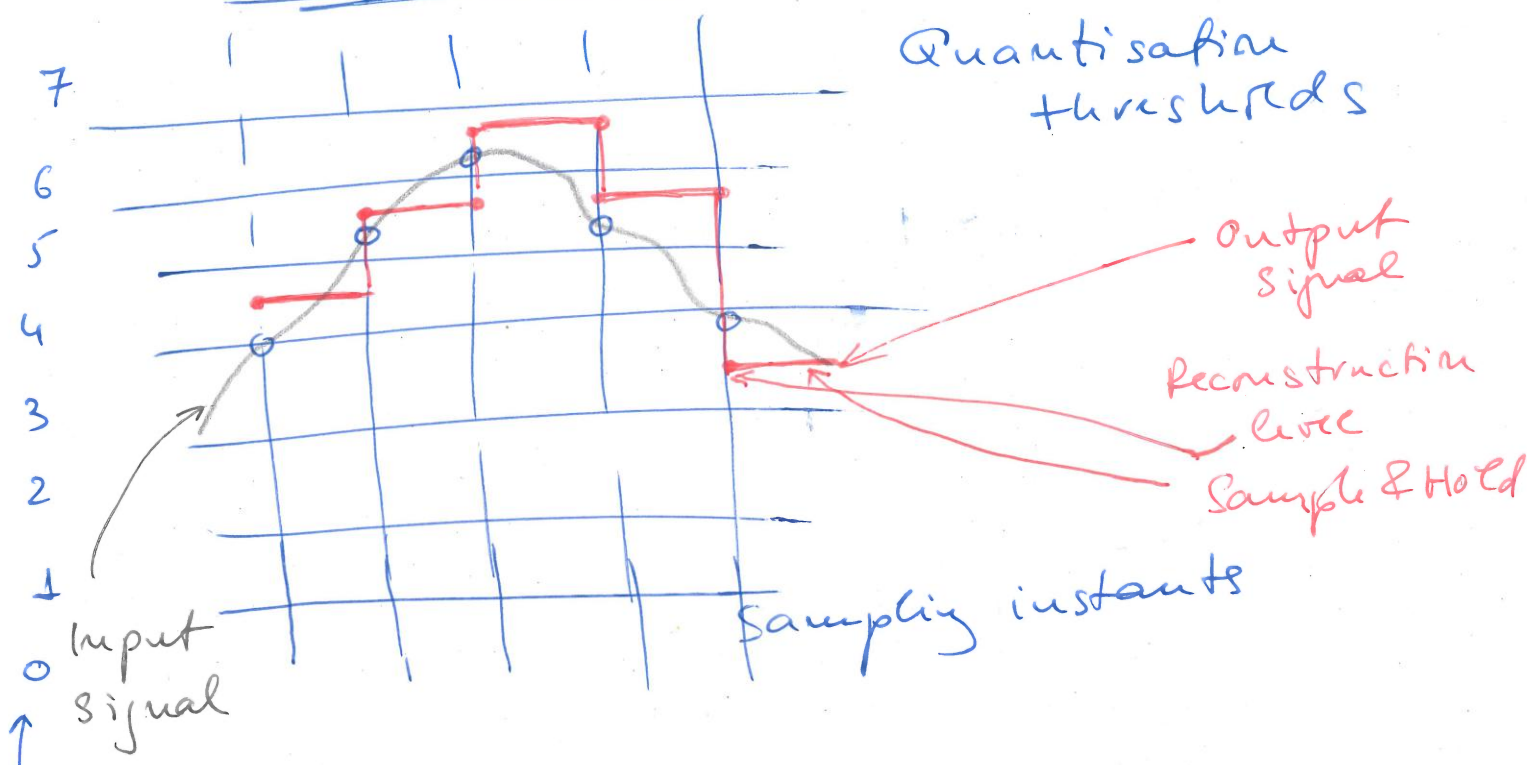
$$\rightarrow F_S = 40 \text{ kHz} < 2F_{MAX} = 2 \times 25 \text{ kHz} = 50 \text{ kHz}$$

There will be aliasing



A/D conversion

Snap to grid



3 bit representation

Samples 4 5 6 5 3

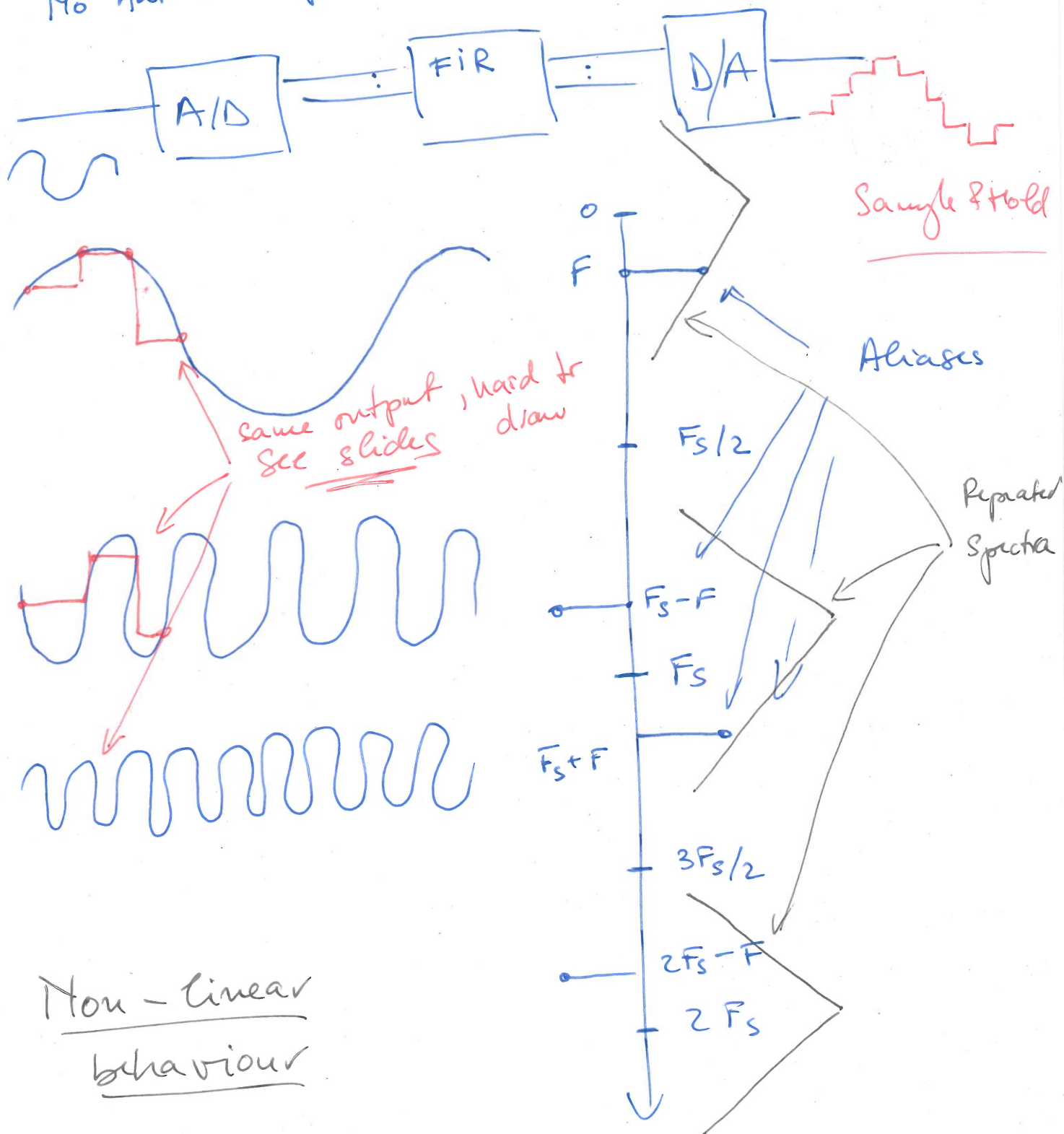
100 101 110 101 011

3 bit integer

Info on lab experiment

8

No Anti-alias filter to start with



Non-linear
behaviour

⇒ Anti aliasing
(Remove input frequencies
 $\geq F_s/2$)

FREQUENCY